CHAPTER

20

Glass-Ionomer Cement

Glass-ionomer cement, developed by Wilson and Kent (1972), was popularized as ASPA (aluminosilicate polyacrylate) in early years.

The glass-ionomer cement was evolved as a hybrid from silicate and the polycarboxylate cement; mainly to overcome deficiencies of silicate cement. The term glass-ionomer cement was coined by Wilson and Kent, though the ISO terminology for the cement was polyalkenoate cement (Fig. 20.1). Because of the extensive use of this cement as a dentin replacement material, it has also been referred to as '*man made dentin*' and '*dentin substitute*'. The greatest advantage of this cement was its adhesion to enamel and dentin and the fluoride release for anticariogenic effect. The most useful clinical applications of glass-ionomer cements have been in restoring erosion lesions without cavity preparation and sealing of pit and fissure defects. Though, glass-ionomer was originally aimed to be a restorative material, soon it was modified as luting cement also. The composition of the powder particle, particularly the type of glass, could be altered to achieve different characteristics. The early cement was slow setting and highly technique sensitive. Over the years, it has undergone many changes. The inclusion of tartaric acid, maleic acid, itaconic acid, etc. decreases the viscosity of the liquid polyacrylic acid. There are variety of glass-ionomer cements, each designed and developed for different clinical applications, such as restoration, luting, base-liner, core build-up, geriatric and pedodontic restorations, ART (atraumatic restorative treatment), cementation of orthodontic bands and brackets, etc.

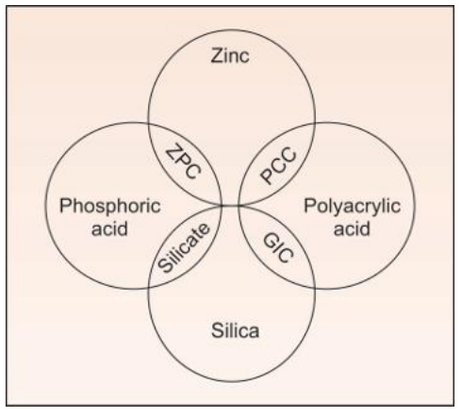


Fig. 20.1: Evolution of glass-ionomer cement. (ZPC—zinc phosphate cement, PCC—polycarboxylate cement, GIC—glass-ionomer cement.)

COMPOSITION

The powder contains silica, alumina and fluorides of calcium, sodium and aluminum. Fluorides are used as flux. The mixture is fused at high temperature of about 1,100 to 1,500°C and the molten mass is shock cooled. This mass is finely powdered to required particle sizes. Restorative cements have a particle size of about 50 μm; whereas, it is about 20 μm for luting cements.

Traditional Glass-Ionomer Cement (GIC)

Powder

Constituents	*% by weight*
Silica (SiO_2)	35–50
Alumina (Al_2O_3)	20–30
Aluminum fluoride (AlF_3)	1.5–2.5
Calcium fluoride (CaF_2)	15–20
Sodium fluoride (NaF)	3.0–6.0
Aluminum phosphate ($AlPO_4$)	4.0–12
Lanthanum, strontium, barium (for radiopacity) (fluorides act as ceramic flux)	In traces

The powder of traditional glass-ionomer cement is a calcium-fluoro-alumino-silicate glass (SiO_2-Al_2O_3-CaF_2-Na_3AlF_6-$AlPO_4$), commonly referred to as '*ion leachable glass*'.

The components of powder are modified to improve upon the mechanical properties and also the area of clinical use.

Some of the modifications of the powder are:

- Dried polyacrylic acid (anhydrous GIC).

- Silver-tin alloy (Miracle Mix).
- Silver-palladium/titanium (Cermet cement).
- BisGMA, TEGDMA and HEMA (light/dual cure GIC).

Liquid

• Polyacrylic acid	45%
• Itaconic acid • Maleic acid	5% (decreases viscosity)
• Tricarballylic acid	
• Tartaric acid	Traces (increases working time and decreases setting time)
• Water	50% (hydrates reaction product)

The liquid is 45% solution of copolymer of polyacrylic acid, 5.0% modifiers and 50% water. The modifiers constitute one or more of the following acids: Itaconic acid, tartaric acid, maleic acid, tricarballylic acid.

The itaconic acid copolymer reduces the viscosity of the liquid. It also inhibits the gelation caused by the intermolecular hydrogen bonding. As a result, it increases the shelf life of the liquid. Tartaric acid is incorporated for ideal manipulative property. It is also a hardener that controls the pH of the cement during setting process, which in turn controls the rate of dissolution of the glass. It facilitates extraction of ions from the glass. It increases the working time and also aids in a snap set (Fig. 20.2).

The components of liquid are also modified for better properties; the modifications are:

- Only water and tartaric acid (anhydrous cement).
- HEMA (light cure components).

CLASSIFICATION

A. According to Wilson and McLean

1. Type I—luting cements
2. Type II—restorative cements
   - Restorative esthetic
   - Restorative reinforced

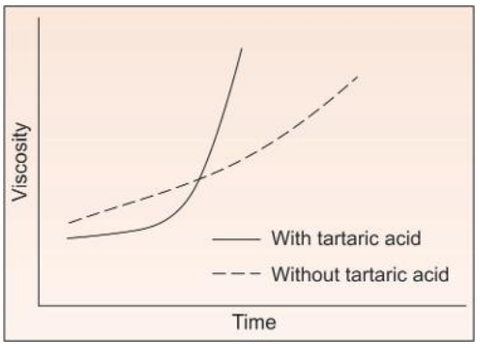


Fig. 20.2: Effect of tartaric acid on the viscosity of the glass-ionomer cement.

B. According to McLean

1. Glass-ionomer cements (traditional)
2. Resin-modified glass-ionomer cements
3. Poly-acid-modified composite resins

C. According to Application

1. Type I—luting cements
2. Type II—restorative cements
   - Esthetic filling materials
   - Reinforced materials (Fuji IX, Fuji II LC)
3. Type III—lining cement
4. Type IV—fissure sealant
5. Type V—orthodontic cement
6. Type VI—core build-up cement

D. According to Characteristics Specified by the Manufacturer

1. Type I—luting cement (Fuji I, Ketac cement)
2. Type II—restorative material (Ketac Fil, Fuji II, Fuji IX, etc.)
3. Type III—base and liners
   a. Weak with low acidity (GC lining cement, Shofu Liner)
   b. Stronger, but more acidic (Ketac Bond, Shofu base, GC dentin cement)
   c. Strong, even in thin layer (light cure—Vitrebond)
4. Type IV-Admixtures (Ketac silver, Miracle mix)

E. Newer Classification

1. Traditional glass-ionomer
   - Type I—luting cement
   - Type II—restorative cements
   - Type III—liners and bases
2. Metal modified glass-ionomer
   - Miracle mix
   - Cermet cement
3. Light cure glass-ionomer (HEMA added to liquid)
4. Hybrid glass-ionomer/resin modified glass-ionomer
   - Composite resin in which fillers are substituted with glass-ionomer particles.
   - Precured glasses blended into composites.

USES OF GLASS-IONOMER CEMENT

1. Restoration of permanent teeth
   - Class V and III cavities
   - Abrasion/erosion lesion
   - Root caries
2. Restoration of deciduous teeth
   - Class I–VI cavities
   - Rampant caries, nursing bottle caries
3. Luting/cementing
   - Metal restorations, viz. inlays, onlays, crowns

- Non-metal restorations, viz. composite inlays and onlays
- Veneers
- Pins and posts
- Orthodontic bands and brackets

4. Preventive restorations
   - Tunnel preparation
   - Pit and fissure sealants
5. Protective liner, under composite and amalgam
6. Bonding agent
7. Dentin substitute
8. Core build-up
9. Splinting of periodontally weak teeth
10. Glazing
    - Glazing of traditional GIC filling
    - Improving esthetics of old GIC filling
    - Protection of new GIC filling
11. Other restorative technique
    - Sandwich technique/layered restorations/laminated restorations/bilayered restorations
    - Atraumatic restorative treatment
    - Bonded restorations
12. Endodontics
    - Repair of external root resorption
    - Repair of perforation
    - Retrograde filling

Advantages (conventional glass-ionomer cement)

- Chemical adhesion
- Releases fluoride (anti-cariogenic)
- Biocompatible
- Low oral solubility (after complete set)
- Easy manipulation

Disadvantages (conventional glass-ionomer cement)

- White and opaque surface
- Clinically poor esthetics
- May exhibit cracks on surface
- Debonds readily
- Highly technique sensitive
- Solubility is high initially (first 24 hours)
- Poor strength
- Not successful in larger lesions

SETTING REACTION

The setting reaction varies in conventional (chemical cure) glass-ionomer, as acid–base reaction and light activated polymerization in resin modified (dual cure) glass-ionomer cements. Tri-cure activation involves reduction-oxidation between initiator and the catalyst.

1. Acid-base reaction
2. Light-activated polymerization
3. Tri-cure activation

1. Acid–Base Reaction

The acid–base reaction involves three components, viz. fluoroaluminosilicate glass powder, an ionic polymer of polyacrylic acid and water. The reaction occurs between glass powder and the ionic polymer. Water is an essential medium through which ion transfer takes place. While mixing the powder and liquid, the acid slowly degrades the outer layer of glass particles (calcium ions and aluminum ions). During the early stages, divalent calcium ions are released more rapidly and are primarily responsible for reacting with the polyacid to form a reaction product, i.e. calcium polysalts. The initial set is due to this reaction. At this stage, the relationship with water is critical. Hence, the cement is to be protected from coming in contact with external moisture as well as drying. If the cement comes in contact with moisture, the aluminum ions may leach out of cement, before it combines with the main network. This will result in a weak cement as well as a more soluble cement, which will be rapidly eroded by the oral fluids. If the cement is dried, then the water, which is essential for ion transfer, will prevent the reaction to completion; resulting in weak material, which may crumble easily. It also appears dull and white and exhibits poor esthetics. Trivalent aluminum ions are released more slowly and become involved in setting at a later stage often referred to as '*secondary reaction stage*'. The aluminum ions also replace divalent calcium ions and form tight network of crosslink between polymer chains. The set material consists of unreacted glass cores embedded in a matrix of cross-linked polyacid (Fig. 20.3). The second stage of setting reaction involves the incorporation of significant quantities of aluminum in the matrix structure resulting in adequate physical properties. This stage requires about 24 hours. Although the material may appear to be hard within the first 5 minutes, it takes at least 24 hours before the material is set to a stable form.

For a complete reaction to take place, it may take as long as 7 days (Fig. 20.4). Once the replacement reaction is complete, the solubility of the cement in water also decreases. Silica ions react with the available water

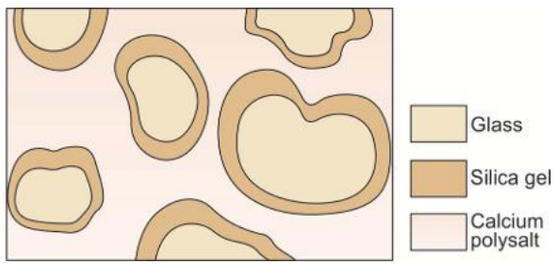


Fig. 20.3: Diagrammatic representation of final set cement

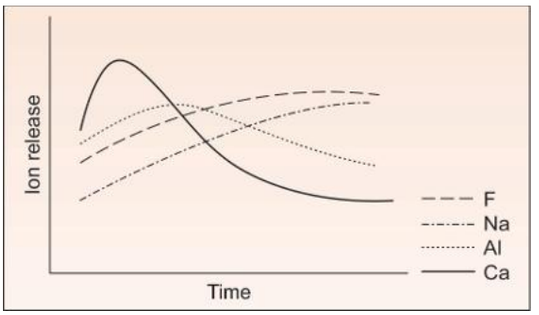


Fig. 20.4: The effect of time during setting on the release of different ions

and form covalent silica network around the glass particles. The fluoride in the cement exists in a dynamic relationship with the oral environment. There is a continuous leach of fluoride throughout the lifetime of the cement. The initial release is high followed by a gradual decrease to reach a constant level; however, whenever there is an increase in the fluoride level in the environment, the cement imbibes the lost fluoride and stores it like a reservoir to release gradually over a period of time.

The above reaction is also described in three overlapping stages, viz. dissolution, gelation and hardening.

2. Light-Activated Polymerization

The setting reaction in light cure glass-ionomer is basically a two-stage process. Since the powder component of light cure glass-ionomer cement consists of initiators for light curing and the liquid component is modified with hydroxyethyl-methacrylate (HEMA), these ingredients are responsible for polymerization by light. The initial reaction leads to the polymerization of methacrylate groups and subsequently followed by acid–base reactions of the glass component of powder and polyacrylic acid of the liquid. This is also known as *'dual cure cement'*.

3. Tri-Cure Activation

In addition to the light curing mechanism and the acid–base reaction, some of the resin-modified glass-ionomer cements harden by a chemical 'dark cure' catalysed by a reduction–oxidation reaction. The dark cure is perhaps important in deeper restorations in which light beam penetration may not be sufficient for completion of the light-hardening phase and chemical reaction is also insufficient. These materials are termed *'tri-cure'* as these resin modified glass-ionomer cements set by three phenomena.

i. Acid–base reaction between the conventional components
ii. Light-cure reaction stimulated by light application.
iii. Autocure reaction, which is a reduction–oxidation (Red-Ox) reaction that takes place between the initiator-catalyst system of resin that gets activated when the powder and liquid components are mixed together. This reaction ensures that with time, there will be a complete cure of the entire restoration with no free-resin remaining.

PHYSICAL AND MECHANICAL PROPERTIES

The working time of traditional glass-ionomer cement is 2–3 minutes. The water settable systems tend to have longer working time. Solubility in water during the first 24 hours is high. Glass-ionomer has a co-efficient of thermal expansion close to that of dentin. It inhibits bacterial growth, due to fluoride release by the cement. The glass-ionomer cement is a mild irritant to pulp tissue, especially when placed over dentin. The biocompatibility of the cement is due to the large molecular size of the cement, which does not travel through the narrower dentinal tubules. The initial pH of the mixed cement is close to 2 but soon reaches a value between 6 and 7. However, glass-ionomer should not be placed over a direct exposure of the pulp. In case, where the remaining dentin is less than 0.5 mm, it is advisable to protect dentin and the pulp with calcium hydroxide.

The physical and mechanical properties of glass-ionomer cement and also the modified forms are summarized in Table 20.1.

OTHER PROPERTIES

1. Sensitivity to Moisture

Glass-ionomer cements are very sensitive to moisture, especially during initial setting phase. Both sorption of water and over drying are deleterious to the mechanical properties of the set cement. Absorption of water lead to weak cement and the over drying will lead to cracks in the cement. Therefore, protection of the cement during and after placement is mandatory.

2. Postoperative Sensitivity

Postoperative sensitivity is usually associated with poor manipulation and/or poor powder/liquid ratio. This is also related with moisture contamination during setting of the cement, leading to hydraulic effect on dentinal fluid. However, light cure glass-ionomer and compomer exhibit less postoperative sensitivity.

3. Adhesion

One of the most important characteristics of glass-ionomer cement is their ability to adhere chemically to mineralized tooth tissues. As the mixed cement is placed, an immediate ion exchange between cement

Table 20.1: Physical and mechanical properties of glass-ionomer cement and its modified forms

	Traditional GIC	*Glass cermet*	*Light cure GIC*	*Compomer*
Compressive strength	16000–22000 psi	27500 psi	23200 psi	29000 psi
Tensile strength	900–960 psi	950–1000 psi	2000–3000 psi	2000–3000 psi
Modulus of elasticity	2900000 psi	–	906175 psi	1217900 psi
Coefficient of thermal expansion/°C	10–11 × 10^{-6}	15 × 10^{-6}	18 × 10^{-6}	18 × 10^{-6}
Solubility	0.3–0.5%	0.1%	0.08%	0.08%
Opacity	90%	95%	85%	80%
Hardness (KHN)	48	39	40	38
Film thickness (μm)	22–25	–	30–40	30–40

and the mineralized tissues takes place. The probable mechanism of adhesion is based on both diffusion and adsorption phenomena. The polyacrylic acid will penetrate the tooth structure, releasing phosphate ions. These phosphate ions take up calcium ion from the tooth surface to maintain electric neutrality. These ions will combine with the surface layer of cement and form an intermediate layer of new material, which is firmly attached to the tooth surface. This has been described as a 'diffusion-based adhesion'. The adhesion to dentin is also through hydrogen bonding or metallic ion bridging between the carboxyl groups of the polyacid and the collagen of dentin.

Summarily, adhesion of glass-ionomer cements can be attributed to two inter-related phenomena:

- Micromechanical (polyacid components of glass-ionomer make it self-etching).
- Chemical bonding (ionic bonds formed between carboxyl group or polyacid molecule and calcium in the tooth surface).

As the cement matures, a diffusion process occurs in which ions from the cement and ions from the tooth move into the interfacial zone and create an ion-exchange layer. This layer causes the tooth and the cement to adhere strongly.

It has been established that failure in glass-ionomer is cohesive in nature (occur within cement, rather at cement–tooth interface). The boning values can differ at different sites of the restorations.

Adhesion aids in retention and reduces marginal leakage. (Harmful micro-organisms cannot enter the tooth–restoration interface.)

4. Fluoride Release and Recharge (Anticariogenicity)

Incorporation of fluoride in glass-ionomer acts as a flux during sintering of the powder particle. Fluoride also prolongs the working time of the cement. Fluoride release in glass-ionomer can continue for years after the cement hardens. There is an initial rise in fluoride release from the cement for about a week and then gradually decreases within a few months. The most advantageous phenomenon of the cement is that it can also take up fluoride from the environment and replenish itself, functioning like a reservoir of fluoride in the oral cavity.

Fluoride release also provides microbial inhibition. Various additives with the ability to enhance the antimicrobial effect have been tried. The addition of zinc sulfate to glass-ionomer has shown significant inhibition of *Streptococcus mutans* growth and also increased fluoride release.

Fluorides reduce the metabolic activity of the cariogenic microflora in the oral cavity.

It converts calcium hydroxyapatite crystals of the dental tissues to acid resistant calcium fluorapatite crystals that resist acid attack from caries producing micro-organisms.

A few authors have questioned the validity of fluoride release and its anticariogenic properties. In one of the surveys, glass-ionomer restorations were repeatedly replaced due to involvement of secondary caries.

Fluoride release is considered beneficial. A continuous supply of low level fluorides can inhibit dentin demineralization. Fluoride release may also reduce dentin hypersensitivity. However, release from glass-ionomer to salivary components has not been authenticated.

Fluoride is also taken up by glass-ionomer cements, mainly in early stages (even fluoride-free glass-ionomers, when exposed to fluorides, exhibited fluoride-releasing activities), commonly referred to as '*fluoride recharge phenomena*'.

This is a new development of a fluoride release material to overcome the shortcomings faced by fluoride-releasing material.

The shortcomings are:

- Greater the fluoride release in a material, more open is the structure, resulting in low strength material.
- In order to improve the strength of fluoride-containing materials, they are made more dense;

however, then the efficacy of fluoride release is decreased. Soon after placement, there is sudden burst of fluoride release, followed by a rapid decline in the release rate.

To overcome these shortcomings, fluoride recharge material was developed.

This modified glass-ionomer is in two parts:

i. Restorative part
ii. Charge part

i. The restorative part implies when the 1st burst of fluoride is expelled, the therapeutic potential of the restoration is over.
ii. The material is given a second fluoride charge by using a gel material—charge part that replenishes the fluoride site in the restoration by ion exchange and recovers the fluoride release and also therapeutic potential of the restoration. This is achieved without replacing the material.

The process of fluoride release is affected by extrinsic and intrinsic factors, viz., powder/liquid ratio, mixing time, temperature, solubility and finishing processes, storage medium and the environmental temperature (extrinsic factors).

The diffusion of fluoride into the material and differences in surface energy may affect fluoride recharge ability. It is hypothesized that the recharge ability is attributed to the structure of hydrogel layer around glass filler particles. It is also established that material with higher fluoride release have higher recharge ability.

Recently, hydroxy apatite nanoparticles (8.0% by wt.), were incorporated to enhance fluoride ion release, mechanical properties and inhibiting residual bacteria around cavity margins (cellulose nanocrystal have also being tried).

5. Bioactvity

Glass-ionomer cement is natural bioactive material, since it releases fluoride, sodium, phosphate (bioactive ions), etc. into the surrounding medium. In acidic conditions, these ions are released in higher quantities. These ions play variety of biological roles in the environments, maintaining the balance in saliva and the mineral phase of the tooth (glass-ionomers do take up ions). Cement may take up calcium and phosphate ions from saliva and develop a harder surface, as it matures.

6. Esthetics

Glass-ionomer cements are manufactured by the addition of pigments (metal oxides) to achieve different shades. Glass-ionomer lacks the translucency to simulate natural enamel. It is inferior to composites and ceramics in this aspect. It is opaque and may become dull and lifeless in course of time. Esthetically, compomers are better than light cure glass-ionomer and traditional glass-ionomers.

MODIFIED GLASS-IONOMER CEMENTS

1. Metal-Modified Glass-Ionomers

i. *Miracle Mix (Silver Cermet)*

Silver cermet was prepared by incorporation of silver-tin alloy into the glass-ionomer cement powder. Glass is generally brittle and addition of silver improves toughness of the cement. (Silver acts as a stress absorber.) It also improves the abrasive resistance of the cement; however, other properties, viz. compressive strength, flexural strength, solubility and abrasive resistance remained without improvement. It is esthetically unacceptable because of its gray/black color (Fig. 20.5). It did not exhibit promising results, due to metal–carboxylate interface failure.

ii. *Glass Cermet*

Glass and metal powder is sintered at high temperature. The mixture is made to react with polyacrylic acids to form improved glass-ionomer cement (glass cermet). The cement exhibit better wear resistance and flexural strength; maintaining the esthetics. Metals, like titanium oxide, have also been added in smaller proportion (approximately 5%) as whitening agent. These cements are preferred in small Class I cavities for restoration of deciduous teeth and for core preparation.

These are widely used in Class V, erosion/abrasion cavities and also in rampant caries. They are well tolerated by the pulp and can be finished and polished immediately after setting. The cement is suitable as restorative material, as luting cement and also as base under different restorative materials.



Fig. 20.5: Miracle mix

Advantages of metal-modified glass-ionomers

- Stronger
- Nearly insoluble
- Fluoride release
- Good radiopacity
- Prevent secondary caries when used as a liner at cervical wall of Class II cavities
- Good biocompatibility

Disadvantages of metal-modified glass-ionomer cement

- Unesthetic (black color)
- Long-term failure (failure of metal–carboxylate interface)
- Used only for smaller lesions

2. Resin-Modified (Light Cure) Glass-Ionomer Cement

A monomer, hydroxyethyl methacrylate (HEMA) is grafted in the polyacrylic acid. With the exposure of light, polymerization is initiated along the methacrylate groups. After that the polyacrylic acid reacts with glass particles though acid–base reaction. As these are water based, the main reaction is the acid–base reaction. They maintain the ability to bond to hard tooth tissues via the carboxylic group of polyalkenoate component. The fluoride release is also optimum. Fuji II LC is one commercial preparation of resin-modified glass-ionomer (Figs 20.6 and 20.7).

3. Compomer (Polyacid-Modified Composite Resin)

Compomer is the combination of composites ('comp') and glass-ionomer ('omer'). Compomer contains dimethacrylate monomer and two carboxylic groups along with ion leachable glass. There is no water in the composition of these materials and the glass particles are partially silanated to ensure some bonding with the matrix. These materials set via free radical



Fig. 20.6: Light cure glass-ionomer cement

Advantages of resin modified (light cure) glass-ionomer cement

- Stronger
- Nearly insoluble
- Fluoride release
- Good radiopacity
- Easy manipulation
- Satisfactory wear resistance
- Better esthetics
- Early resistance to water attack
- Bonding to tooth—chemical and micromechanical
- Bond strength—excellent
- Minimal or no postoperative sensitivity
- Condensable viscosity
- Light/dual cure setting
- Long working time (snap set by photoactivation)
- Can be finished and polished immediately after set
- Repair of fractured restoration possible as the bond between old and new materials is very strong
- Exhibits increased adhesion to composite when used as a base

polymerization reaction and do not bond to hard tooth tissues. There is insignificant acid–base reaction between the unsilanated glass particles and the carboxylic groups.

The physical and mechanical properties of compomers are inferior to hybrid composites.

The bond strengths, are, however, comparable to glass-ionomer cements and resin-modified glass-ionomer cements. The roughness values are also comparable to glass-ionomers and composites.

These materials have been successfully tried in Class V restorations. These are also being used in Class II restorations. A few authors were of the view that the use of conditioner along with compomer



Fig. 20.7: GC Fuji II™ LC: Resin-reinforced glass-ionomer cement

would improve the marginal adaptability as compared to composites. These are also used in bilayered technique (base of compomer, covered with composites).

4. Giomers

A new category of hybrid esthetic restorative material, which differs from both resin-modified GICs and compomers has been introduced, known as 'Giomers'.

Giomers are available as one paste form. They are light polymerizing and require bonding agents for adhesion to tooth structure. The common preparations are: Reactmer (Shofu, Japan) and Beautifil (Shofu, Japan) (Fig. 20.8).

The fluoroaluminosilicate glass in giomers is reacted with polyalkenoic acid and water. This technology differs from compomers, in which a variable amount of dehydrated polyalkenoic acid is incorporated into the resin matrix and the acid does not react with glass until water uptake occurs into the restoration. Giomers do contain glass-ionomer components, but cannot be classified as compomers as the acid–base reaction has already occurred. The term '*prereacted glass-ionomer*' composite is suggested to describe giomers. Like compomers, giomers are light polymerized and require bonding system. (Reactmer bond is a single application bonding agent.) It is glass-ionomer based, all-in-one filled adhesive based on polymer rigid glass technology (consisting of fluoroaluminosilicate glass, acetone, water and initiator). The bonding system is a hydrophilic solution that is extremely effective in wetting the tooth surface in one application.

The etching effect relates to acidic monomers/organic acids, that may interact with the mineral component of tooth substrate and enhance monomer penetration. The adhesive etches and penetrates the etched enamel and hardens on evaporation of the solvent and exposure to light. Giomer-based adhesives create a thick hybrid layer between the restorative and tooth substrate, though their bond strength is lower than two-step bonding system.



Fig. 20.8: Giomers

Indications

- Restoration of root caries
- Non-carious cervical lesions
- Class V cavities
- Caries in deciduous teeth

Reinforced glass-ionomer powder (addition of hollow glass fiber filler partials: 5.0–10% by wt), have shown improved mechanical properties as compared to conventional GIC.

Advantages of giomers

- *Fluoride release:* The giomers release fluoride and help prevention of secondary caries. It has been established that giomer adhesive is more permeable; the permeability in bonded dentin affects the potential of caries inhibition.
- *Fluoride recharging:* The fluoride recharging effect significantly reduces the incidence of recurrent caries.
- *Biocompatibility:* Biologically compatible with pulp.
- *Smooth surface finish and esthetics:* The giomers have better surface finish than conventional GIC and resin-modified GICs. The surface finish is comparable to composites. Esthetically pleasing.
- *Excellent bonding:* The single-step bonding agents provide excellent bonding. The duration of single application of bonding is not crucial even though demineralization of dentin with longer application duration is observed. The bond strength of single application bonding system were comparable to those of compomer, but lower than those of the two-step resin-composites bonding system.
- *Clinical stability:* Clinically stable; giomers perform better than composites. The hardness of polymer-rigid composites was found to be significantly greater than polyacid-modified composites and resin-modified glass-ionomer cements.

Disdvantages of giomers

- Not of much benefit to patients who are at a risk for recurrent caries.
- Giomers exhibit rapid and extensive expansion and should be avoided in tooth preparation involving thin unsupported enamel.

5. Antibacterial Glass-Ionomer Cements

One major cause of failure of glass-ionomer cement is the secondary caries, which implies that fluoride release may not account for adequate antibacterial activity. Initial low pH of the set cement was also not

approved. Over the years, the researchers are trying to enhance the antibacterial activity of glass-ionomer cements.

Various clinical agents have been tried to improve upon the antibacterial effect of the cement without compromising with the adhesion qualities.

The main agents are:

i. ***Chlorhexidine:*** Antimicrobial activity of chlorhexidine against gram-positive bacteria has been accepted. Another feature of chlorhexidine is its impact on fluoride release. (Higher concentration of chlorhexidine increases release of fluoride.) A few studies have observed changes in physical/mechanical properties of glass-ionomer cements with use of high concentration of chlorhexidine (decreases bonding potential and also increases setting time). To overcome the drawback, nominal concentration of chlorhexidine was used. A few authors added chitosan or zeolite nanoparticles to enhance the antibacterial activity.

Recently, finely milled particles of chlorhexidine hexametaphosphate (CHX-HMP) were incorporated in glass-ionomer cement. Long-term release of chlorhexidine, without any adverse effects on mechanical properties, was observed.

ii. ***Triclosan:*** Triclosan is a synthetic and anionic agent with a broad spectrum of antimicrobial activity. 2.5% triclosan added to glass-ionomer has shown better results than chlorhexidine (2.0%). It provides long-term antibacterial activity (gets immobilized with cement particles, controlling triclosan release rate).

iii. ***Quaternary ammonium salt:*** Quaternary ammonium salts disrupt the cell membrane by changing membrane permeability of bacteria.

The long chain (16-carbon substitute) in quaternary ammonium salts shows higher antibacterial activity. But the drawback of such long chain lengths is a resultant decrease in compressive strength of GIC. For achieving proper compressive strength, quaternary ammonium compound with shorter carbon chains (6-carbon) were incorporated into glass-ionomer cements.

Antimicrobial activity of quaternary ammonium salts includes:

- Adsorption to negatively charged bacterial cell surface
- Penetration into cell membrane
- Binding to cell cytoplasmic membrane
- Disturbing the metabolism of cytoplasmic membrane

Antibacterial activity of quaternary salts decreases with human saliva because of electrostatic interactions between quaternary salts and the salivary proteins.

iv. ***Epigallocatechin-3-gallate (EGCG)/Furanone derivatives:*** Epigallocatechin-3-gallate (EGCG) is a potent antioxidant and shown antibacterial activity. However, its antibacterial efficacy is limited to initial 3–4 hours, when incorporated in glass-ionomer cement. After 24 hours, biofilm formation was normal around the cement restorations.

EGCG enhances human pulp cell proliferation; hence, used mainly in regenerative endodontics.

Furanone derivatives, though antibacterial, could not show promising results when added to glass-ionomer cement.

v. ***Nanoparticles:*** Silver, zinc and titanium dioxide have shown antibacterial activity when incorporated into glass-ionomer cement. The addition of nanoparticles is effective in minimizing the biofilm formation; however, their long-term effect is under research. Moreover, silver particles may impart black color to the cement, which is a major drawback.

vi. ***Bioactive glass:*** Bioactive glass has the ability to raise pH level in an aqueous environment by leaching of ions, viz. sodium, calcium, strontium, etc. The antibacterial activity of bioglass is not effective; prevents bacterial growth for a short-term, but adversely affects the mechanical properties of glass-ionomer cement.

vii. ***Miscellaneous:*** Antibiotics, plant extracts and certain herbs have also been tried to achieve antibacterial efficacy in glass-ionomer cement. Antibiotics, though potent antibacterial, usually have shown negative effect on mechanical properties, especially compressive strength and the bonding ability. Recently, addition of ciprofloxacin (antibiotic) and zinc L-carnosine (mucosal defensive agent) have shown improved antibacterial effects, without altering the mechanical properties. Miswak (*Salvadora persica*) has shown promising results, without disturbing the physical/mechanical properties.

Sodium fusidate (fusidic acid and sodium) has shown effective anti-bacterial properties.

Propolis has also been tried as antibacterial agent in glass-ionomer cement. Its effect was observed initially, but not seen in long-term studies. Even the cytotoxic studies have proved that propolis is not safe.

6. Glass Carbomer

Glass carbomer is a modified glass-ionomer cement, with conventional acid–base reaction. Certain components of glasses are modified and another few added to achieve improved mechanical properties.

The modifications are:

- Glasses are washed with strong acid, so as the calcium is depleted from the surface layer (most calcium lies inside the core particles).
- Silicone oil (polymethylsiloxane) is applied to four hydrogen bonds, improving binding properties of cement.
- A secondary filler (bioactive hydroxyapatite) is added to promote formation of enamel-like feature at the tooth–restoration interface.

Glass carbomer can be mixed at high powder:liquid ratio. (Silicone oil adsorb onto the surface of glass and interfere with reaction of glass and polyacid.) The glasses in glass carbomer contain strontium and high amount of silicon. Aluminum, phosphorus, fluoride, etc. are of same quantity as in conventional glass-ionomer.

The material exhibits improved physical/mechanical properties; however, long-term studies could not approve its superior efficacy.

7. Zirconomer (Bioactive Nanoceramics)

Zirconium oxide nanoparticles have been induced into glass-ionomer powder to enhance the strength and esthetics of glass-ionomer cement. As compared to conventional GIC, zirconomer has the better color stability. It has been observed that addition of more than 12% (vol.) of hydroxyapatite/zirconium oxide may deteriorate the properties of the cement. Zirconomer has certain drawbacks, as poor working consistency, longer setting time and rough surface texture.

Other nanoparticles, such as nano-hydroxyapatite crystals, nano-titanium oxide, nano-calcium fluoride, etc., have been incorporated into glass-ionomer powder to improve upon the mechanical properties of the cement. However, various studies have reported cracks in the surface, mainly due to weak zirconium–glass interface. The addition of nanoparticles has limited advantages in the long-term evaluation of glass-ionomer cement.

8. Low pH Smart Materials

Low pH material is based on that fact that fluoride release should be at a low pH, i.e. when caries attack is more threatening. These are developed to release fluoride at a low oral pH. Hence, they are known as 'smart materials'. Fluoride is not released all the time, the episodic release prolongs usefulness of the material.

An overview of antibacterial agents used with glass-ionomer cements is summarized in Table 20.2.

PLACEMENT OF GLASS-IONOMER CEMENT

The clinical placement of glass-ionomer cement involves the following steps:

I. Isolation

Table 20.2: An overview of antibacterial agents used and their effects on glass-ionomer cement

Antibacterial agent	*Type of glass-ionomer*	*Findings: Implications*
Chlorhexidine	• Conventional glass-ionomer • Resin-modified glass-ionomer	• 1.25% chlorhexidine enhances the antibacterial activities. • The antimicrobial activity of chlorhexidine against *Streptococcus mutans* and *Lactobacillus acidophilus* has been established. • Chlorhexidine may lead to decrease in fluoride release and antibacterial effect with time. • The concentration of chlorhexidine may affect mechanical properties of glass-ionomer. • Decreases biofilm formation with potent streptococci.
Epigallocatechin-3-gallate (EGCG)	• Conventional glass-ionomer	• EGCG improves antibacterial activity without effect on fluoride release. • Improve mechanical properties
Furanone	• Light cure glass-ionomer	• Antibacterial activity against *Streptococcus mutans* and *Lactobacillus spp.*
Triclosan	• Conventional glass-ionomer	• 2.5% Triclosan is effective against *Streptococci* and *Lactobacillus*.
Propolis	• Conventional glass-ionomer	• Facilitate inhibition of *Streptococcus mutans* growth without changing the mechanical properties
Nanoparticles (silver and titanium dioxide)	• Conventional glass-ionomer • Resin-modified glass-ionomer	• Potent antibacterial activity • Improve mechanical properties
Bioglass	• Conventional glass-ionomer	• Significantly decreases bacterial colonies • No effect on mechanical properties

II. Tooth preparation
III. Manipulation of cement
IV. Restoration
V. Finishing and polishing
VI. Surface protection

I. Isolation

Glass-ionomer cement is very sensitive for water loss as well as contamination. Saliva, sulcular fluid and gingival hemorrhage have to be controlled during the restoration procedure. Rubber dam, retraction cords and cotton rolls with saliva ejectors are mandatory in the restoration of glass-ionomer, especially, lesions close to the gingival margins of the tooth. Details of isolation devices are described in Chapter 11.

II. Tooth Preparation

Tooth preparation for glass-ionomer restoration may be divided into:
A. Mechanical preparation
B. Surface treatment (conditioning).

A. *Mechanical Preparation*

In Class III cavities, the access is gained through the lingual marginal ridge using small-sized inverted cone bur, the bur is extended towards incisal and gingival area depending upon the extent of caries. The labial enamel is preserved as far as possible (Fig. 20.9A, B).

In Class V cavities, the lesion can only be in enamel or both in enamel and cementum. The cavity preparation follows the access opening through the lesion and extending towards the occlusal, gingival, mesial and distal walls. In case the extent of lesion is too wide, a groove on the occlusal/gingival wall can be given. Similar grooves on all the four sides can be given, if lesion so dictates (Fig. 20.10A to D).

The outline form is dictated primarily by caries, discoloration and esthetics only.

In proximal lesions, no attempt should be made to break the contact to bring the restoration to a self-cleansing area. The unaffected tooth material should be retained to keep as much contact as possible, even if it is not well supported by the dentin.

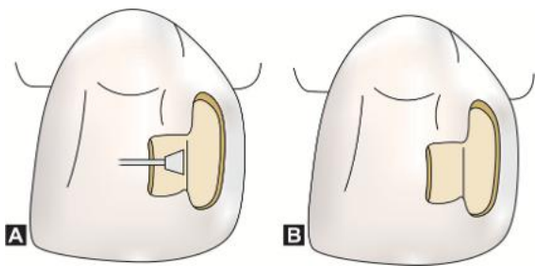


Fig. 20.9: Class III cavity (diagrammatic): **(A)** Access through lingual marginal ridge; **(B)** Prepared cavity

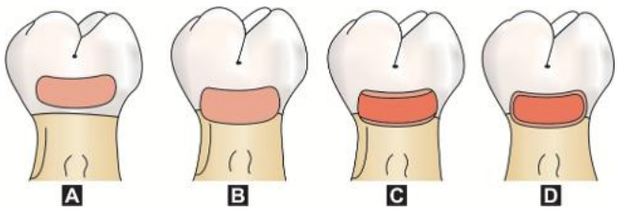


Fig. 20.10: Class V cavity (diagrammatic): **(A)** Cervical cavity only in enamel; **(B)** Cervical cavity both in enamel and cementum; **(C)** Grooves on the occlusal and gingival wall; **(D)** Grooves on all the four walls

The gingival extent of the tooth preparation is generally dictated by the extent of the lesion. It is often close to or extending into the sulcus. Care should be taken to manage the sulcular fluid during placement of the restoration.

B. *Surface Treatment (Conditioning)*

After a thorough prophylaxis, the surface is conditioned using appropriate chemical agent. 50% citric acid, 25% tannic acid and 20% polyacrylic acid are some of the conditioning agents suggested for better bonding of the cement to enamel and dentin. Polyacrylic acid is commonly applied for 10 to 20 seconds and then washed away.

Light-activated glass-ionomers have an additional step of priming the tooth surface. The priming agent is applied in two or more coats as suggested by the manufacturers. It is then spread to a thin layer with gentle blast of air and light activated for 20 seconds.

III. Manipulation of Cement

A. *Dispensing Powder/Liquid*

When dispensing the powder, shake the bottle lightly first to fluff up the powder. See that the spoon is quite full of powder and level off the surface using the lip on the bottle, which is provided for the purpose.

The powder is dispensed first onto the slab and divided into two halves with the spatula. The liquid bottle is then tilted horizontal to allow the liquid to displace the air through the nozzle and occupy the nozzle. The bottle is then oriented vertically and gently squeezed to dispense the required ratio of liquid without the inclusion of air bubbles (Fig. 20.11A to C). If the liquid has become more viscous over time, it can be thinned down by immersing the bottle in water at 70°C for 15 minutes. Allow it to cool again before using the liquid.



Fig. 20.11A: Glass-ionomer (powder/liquid)



Fig. 20.11B: Dispensing of powder



Fig. 20.11C: Dispensing of liquid



Fig. 20.11D: Mixing powder and liquid

B. *Mixing*

The cement is mixed either with hand or using syringe technique (capsulated cement). The mixing technique should be followed according to the manufacturer instructions.

Mixing can be carried out on a paper pad or a glass slab (Fig. 20.11D). The advantage of a glass slab is that it can be cooled a little to prolong the working time. The primary objective of mixing the cement is to merely wet the surface of the powder particles with the liquid and not to aid in dissolving them entirely in the liquid.

As powder and the liquid are dispensed on the pad, one-half of the powder is mixed at a time by rolling the powder into the liquid to wet the surface of the powder particles. Do not vigorously spatulate. (Plastic spatula is preferred over stainless steel spatula.) The first half of the powder should be all wet within 10 to 15 seconds. Roll the remaining powder into the mass. Do not spread the mix around the pad. The mixing is completed in about 30 seconds.

C. *Slump Test*

This test helps to recognize, if cement is properly mixed. Lift the top of the pile of the cement with a spatula. The cement breaks and slumps back. There will be a point when the cement breaks but does not slump. Working time is normally 60–90 seconds for traditional glass-ionomer and about 3 minutes for resin-modified glass-ionomer.

IV. Restoration

After mixing, both the chemical cure cement and the light cure cement are carried in one bulk for placement into the cavity. The use of a matrix is desirable because it assists in positive placement of the cement onto the tooth surface and also leads to reduction of voids and porosities within the restoration.

A syringe dispenser, like the Centrix syringe, can be preferred. Syringing out the cement into the prepared cavity helps to fill the cavity without entrapment of air bubbles in the cement and gives absolute control of quantity of cement placed in the tooth. The tip of the syringe is placed close to floor of the cavity and the cement ejected into the cavity as the nozzle is withdrawn out. A light cure restoration should be subjected to longer exposure than suggested to avoid deficient light activation. A fully light activated resin- modified glass-ionomer is resistant to water uptake after completion of light activation, but it is still possible to dehydrate the cement.

V. Finishing and Polishing

The best finish is necessarily the matrix finish. Initial contouring may be carried out using sharp instruments, like gold foil knives/diamond points. Many clinicians prefer initial contouring after 24 hours to avoid water from affecting the cement. The final finishing is carried out using 'Sof-lex' discs. Super fine diamond points, silicon abrasives embedded in rubber in various shapes and abrasive strips are available to finish various areas of the restorations (Fig. 20.12). The finishing has to be carried out in moist condition. Dry finishing may dehydrate the surface, leading to porous and mottled appearance; subsequently, early disintegration of the cement. At the end of the initial contouring as well as final finishing of the surface, the cement surface is preferably protected with resinous glaze material (described in detail in Chapter 23). The scanning electron microscopic picture of finished surface of conventional glass-ionomer is shown in Fig. 20.13A to C and light cure glass-ionomer is shown in Fig. 20.14A to C. Final polishing is delayed for 24 hours (details in Chapter 23).

VI. Surface Protection

Earlier, varnish was used to protect cement surface; however, resins, such as enamel bonding agents, afford the best surface protection to the cement. Resin material fills the irregularities on the surface and gives a smooth finish for a longer period as compared to varnish and petroleum jelly. Resins are also more impermeable

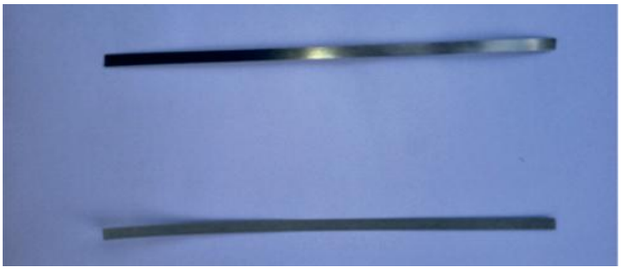


Fig. 20.12: Metal finishing strips for glass-ionomer restorations

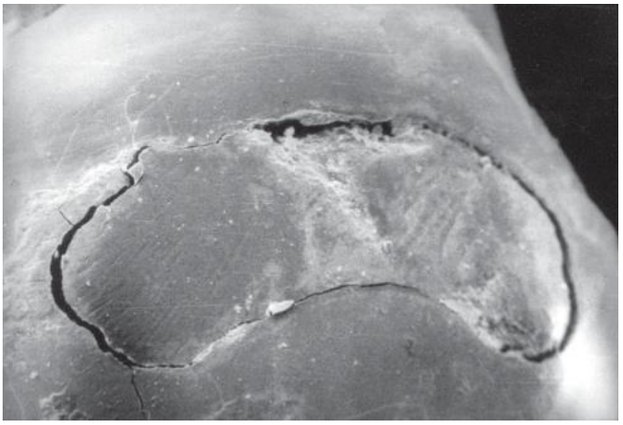


Fig. 20.13A: SEM photograph of finished conventional GIC (magnification 25x)

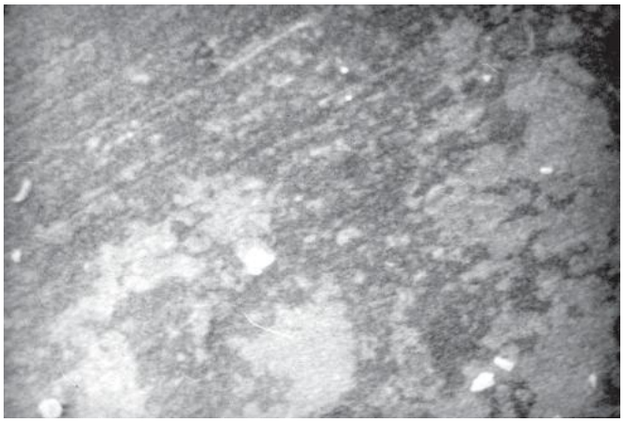


Fig. 20.13B: SEM photograph of finished conventional GIC (magnification 200x)

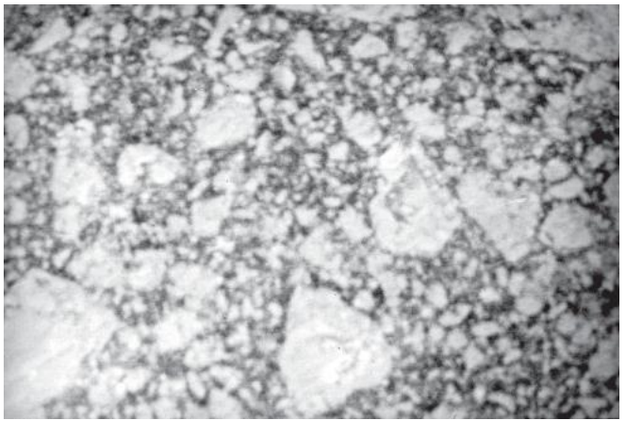


Fig. 20.13C: SEM photograph of finished conventional GIC (magnification 1000x)

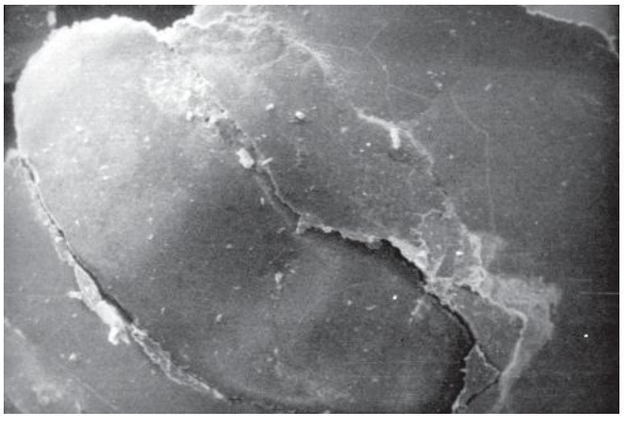


Fig. 20.14A: SEM photograph of finished light cure GIC (magnification 25x)

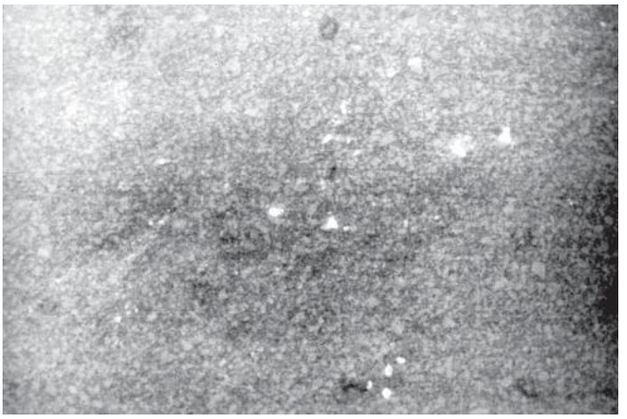


Fig. 20.14B: SEM photograph of finished light cure GIC (magnification 200x)

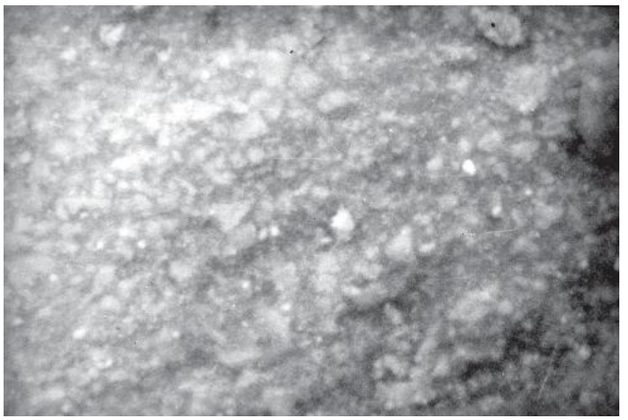


Fig. 20.14C: SEM photograph of finished light cure GIC (magnification 1000x)

compared to varnish. Cocoa butter and petroleum jelly are easily washed away in a short period while optimum period of protection is 24 hours.

The newly placed cement should be sealed immediately after removal of the matrix to prevent water exchange. If the seal remains for the first 24 hours, the cement would mature sufficiently to develop full translucency. In case, the surface is not protected during maturation, it will lead to crack formation at the surface (Fig. 20.15). The problem of developing full translucency and esthetics was solved with the use of a very low viscosity, light-activated, resin enamel bond as a sealant, rather than the recommended varnish, which is likely to allow some transmission of water.

CLINICAL APPLICATIONS

a. Restorative

i. *Abrasion–Erosion Lesion*

In Fig. 20.16A, note the diffuse outline at the cervical portion and the well-defined margin at the occlusal extent with involvement more towards the distal of the root surface. The defect is a combination of abrasion

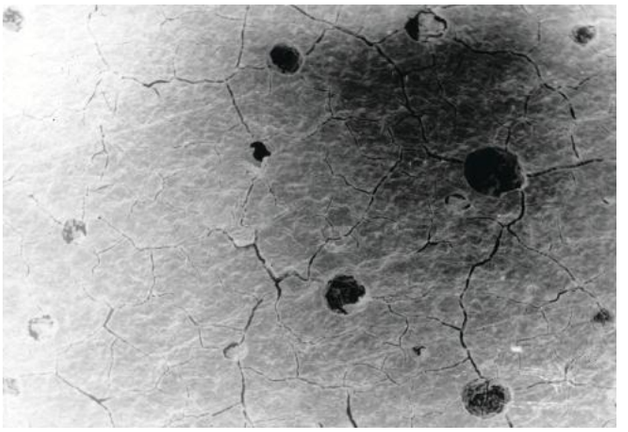


Fig. 20.15: Cracks at the surface of glass-ionomer restoration

(notch-like deep defect at the occlusal margin) and erosion (surface is also smooth and glossy) and also abfraction as the lesion extends into the sulcus, where the cementoenamel junction was present. The lesion was restored with glass-ionomer cement. The ion exchange adhesion of glass-ionomer is superior and the cement appears to be sufficiently flexible to with stand the stresses of flexion in these teeth (Fig. 20.16B).



Fig. 20.16A: Cervical lesions on maxillary canine and first premolar



Fig. 20.16B: Restored lesions



Fig. 20.17A: Abrasion–erosion lesions with caries (maxillary canine and first premolar)



Fig. 20.17B: Restored lesions

ii. *Abrasion–Erosion Lesion with Caries*

Figure 20.17A shows erosion–abrasion lesions in maxillary canine and first premolar with caries superadded. The premolar has a 'V' shaped and notched defect that is typical of an abrasion lesion. (It could have arisen through 'abfraction' also.) The lesion was restored with glass-ionomer cement (Fig. 20.17B).

iii. *Abfraction Lesions*

Abnormal physical forces, many a times, lead to microcracks at the cementoenamel junction. These cracks subsequently help in development of caries. In case the forces remain active for a longer period, the lesions become wider and deeper. The multiple abfraction lesions in the maxillary teeth are restored with glass-ionomer cement (Fig. 20.18A,B).

iv. *Class V Cavities*

An atypical cervical carious defect is seen in Fig. 20.19A. The occurrence does not fit into what can be termed as 'rampant' caries. If a root surface lesion on the exposed root of a tooth is found, the remainder of the dentition should be carefully explored for other root surface lesions, because they generally do not occur in isolation.



Fig. 20.18A: Multiple abfraction lesions



Fig. 20.18B: Restored lesions



Fig. 20.19A: Cervical caries (preoperative)

The lesions were restored with glass-ionomer cement (Fig. 20.19B to D).

v. *Class III Cavities*

Class III cavities (Fig. 20.20A) are best restored with light cure glass-ionomer for better esthetics than with conventional chemical cure cement. Only hidden



Fig. 20.19B: Placement of cord



Fig. 20.19C: Tooth with gingival cord in place



Fig. 20.19D: Restored lesion

lesions may be restored with conventional cement. The auto-cure cements have superior fluoride release than the light cure cement. A 'T' shaped matrix is fabricated as shown in Fig. 20.20A, B. The cavity is restored while the strip is in place (Fig. 20.20C).

The restoration is light activated through the matrix for 20 seconds, then the matrix is removed and cured further (Fig. 20.20D).



Fig. 20.20A: Class III lesion in maxillary lateral incisor



Fig. 20.20B: Mylar strip in place ('T' shaped)



Fig. 20.20C: Holding the strip tight after insertion of the restorative material



Fig. 20.20D: Finished restoration



Fig. 20.21A: Prepared Class I cavity



Fig. 20.21B: Finished restoration

vi. *Class I Cavities*

In conservative Class I preparations, glass-ionomer cement can be preferred, especially in individuals with high caries indices (Fig. 20.21A, B).

b. Lining

Glass-ionomer cement is used as lining cement. A lining is a thin layer placed under a metallic restoration, which is intended to afford thermal protection to the pulp-dentin organ. It is also used to fill up undercuts in cavity design where a line of withdrawal is required as in an inlay cavity or crown preparation. The type-III lining cement has limited physical properties and should not be relied upon as a base for a composite resin in the lamination technique. (Type-II restorative or a light-cure cement is preferred.)

c. Luting

Glass-ionomer cement is used for luting crowns and bridges, as well as orthodontic appliances. The advantages are its anticariogenic property, low solubility in oral environment, pulp compatibility, good thixotropic flow, low film thickness, and also the adhesive property and tooth color.

The tensile strength of the cement is not sufficient to enhance retention. To ensure adequate flow properties, the cement is mixed at a low powder to liquid ratio (does not possess adequate physical properties).

The luting cements are fast setting and can, therefore, be exposed to water contamination within 5 minutes of the start of mix.

The correct mix of the cement for luting is demonstrated by the string test. When mixed, it should be able to string out from the pad to about 3.0 cm or before it breaks and slumps back into the body of the mix. A small quantity of the cement is painted on the prepared tooth surface, using a small bristle brush, to minimize the chances of air inclusions under the crown. Then coat or fill the restoration with the cement and seat it in place, applying occlusal forces.

Glass-ionomer does not have an elastic memory and the restoration will not retreat from the tooth as happens with zinc phosphate cement.

The excess cement may be removed immediately upon initial set with a sharp instrument. These luting cements are radiopaque; any proximal residue can be detected with a radiograph.

Some patients experience severe sensitivity on seating of the restoration with this cement. The reason attributed is the hydraulic displacement of dentinal fluid. Hence, it is desirable to seal the dentin tubules by painting the prepared surface with a low viscosity resin enamel bond or apply a remineralizing solution.

d. Bilayered/Sandwich Restorations

Traditonal type-II and light-cure glass-ionomer cements are used as base, over which composite restoration is placed (sandwich restoration). The comparatively better adhesion of the cement allows it to bond to the dentin in erosive lesions, leaving out small area of enamel that can be used for bonding composites. It is also used for bonding dental amalgam to posterior teeth where there is compromised retention form due to loss of walls to support the amalgam. Glass-ionomer is preferred as a base in Class II lesions, where cervical extension is subgingival. A glass-ionomer is easier to place in these locations than the technique sensitive composite (Fig. 20.22A to D).

FAILURE OF GLASS-IONOMER RESTORATION

The failure of glass-ionomer restoration can be because of the following reasons:

- Dissolution of the immature cement
- Long-term erosion
- Loss of cement by abrasion
- Fracture of restoration

Dissolution of some amount of cement occurs from the surface during first 24 hours, because the surface is not sufficiently matured. Temporary protection with a layer of nitrocellulose, methyl methacrylate or

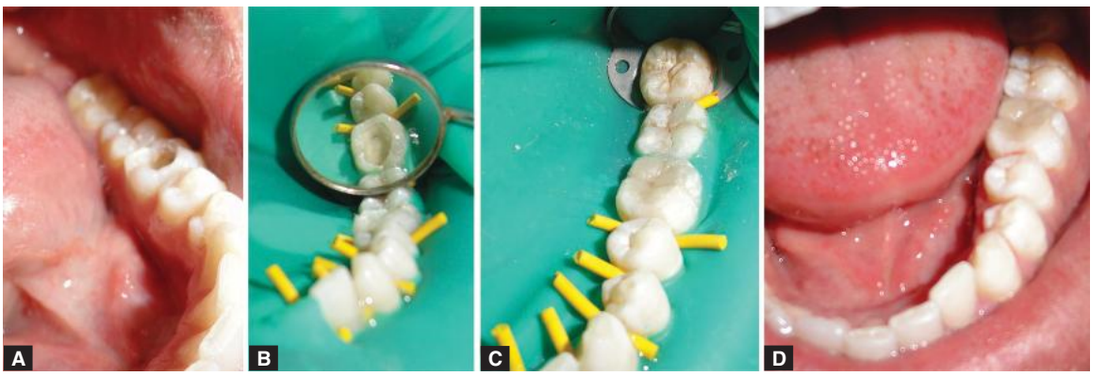


Fig. 20.22: Bilayered technique: **(A)** Prepared cavity for bilayered technique; **(B)** Base of glass-ionomer cement; **(C)** Cavity restored with composite; **(D)** Finished restoration

amide resins in the form of varnish applied over the restoration will help prevent this loss to a great extent. Cocoa butter and vaseline also help, though to a lesser extent. Long-term erosion and loss of cement by abrasion is expected, especially if the restoration falls under susceptible areas. The restoration may fracture as a result of flexural change in the tooth. The cervical areas of the teeth, which undergo maximum flexural change, are likely to dislodge the restoration. It is also observed that the fracture often occurs within the body of the restoration (cohesive failure). The cement often leaves an acid-resistant hybrid surface layer when it fractures from the defective area.

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